

Day 16 - (VLANs pt 1)

Saturday, August 22, 2026

2:50 PM

16. VLANS : PART 1

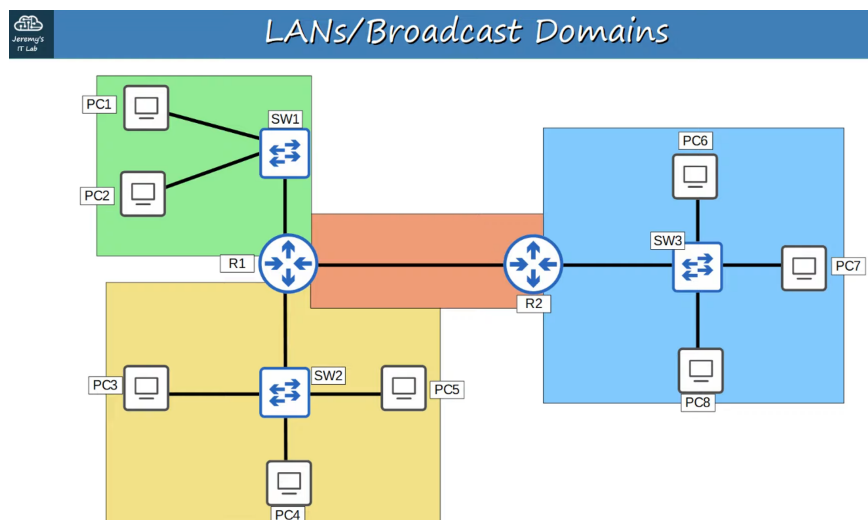
WHAT IS A LAN ?

- A LAN is a single BROADCAST DOMAIN, including all devices in that broadcast domain.

BROADCAST DOMAINS

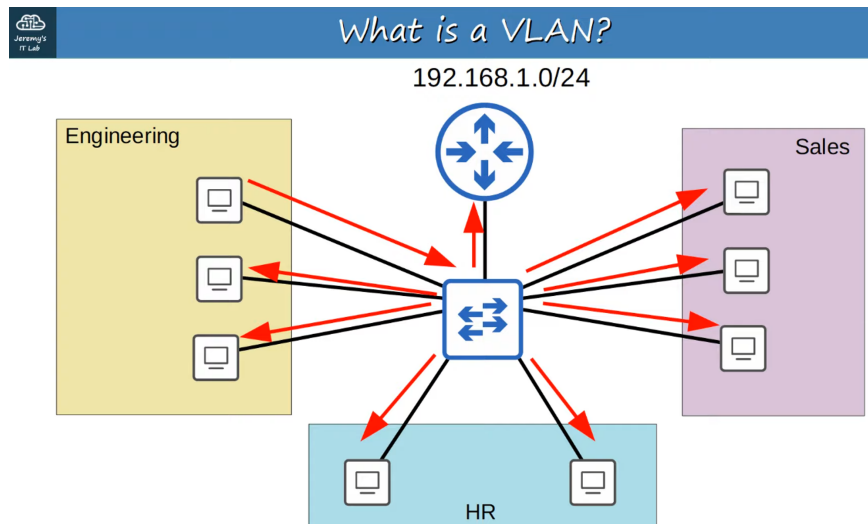
- A BROADCAST DOMAIN is the group of devices which will receive a BROADCAST FRAME (Destination MAC : FFFF.FFFF.FFFF) sent by any one of the members.

Image of LAN with FOUR BROADCAST DOMAINS (192.168.1.0 / 24)

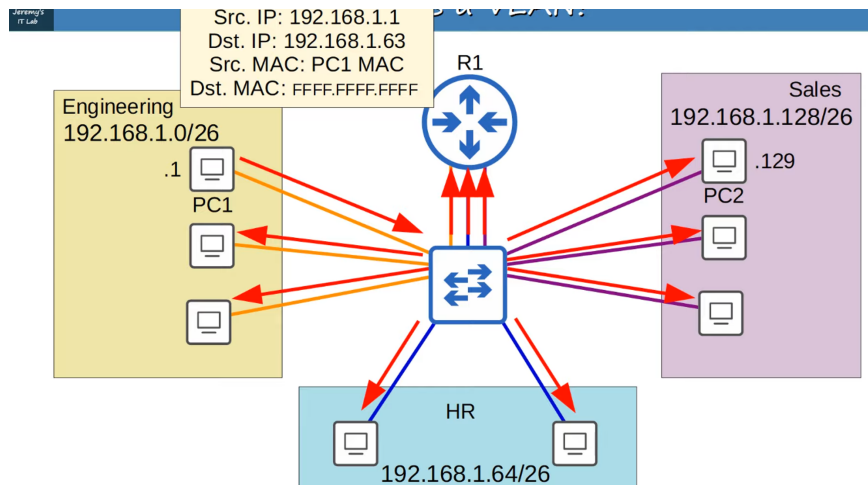


Performance :

Lots of unnecessary BROADCAST traffic can reduce network performance.

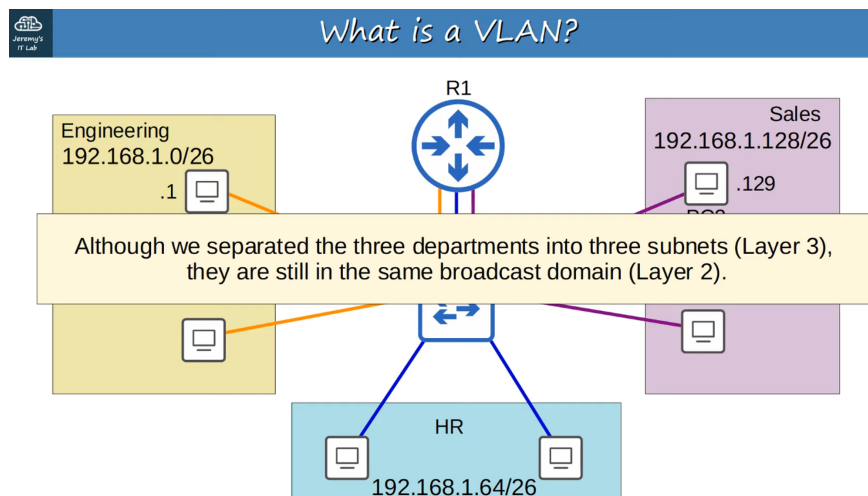


BROADCAST FRAME flooding all our subnets with unnecessary traffic.



Security :

Even within the same office, you want to limit who has access to what. You can apply security policies on a ROUTER / FIREWALL. Because this is one LAN, PC's can reach each other directly, without traffic passing through the router. So, even if you configure security policies, they won't have any effect.



WHAT IS A VLAN ?

VLANS:

- logically separate end-hosts at LAYER 2
- are configured on Layer 2 SWITCHES on a per-interface basis.
- any END HOST connected to that interface is part of that VLAN

PURPOSE OF VLANS:

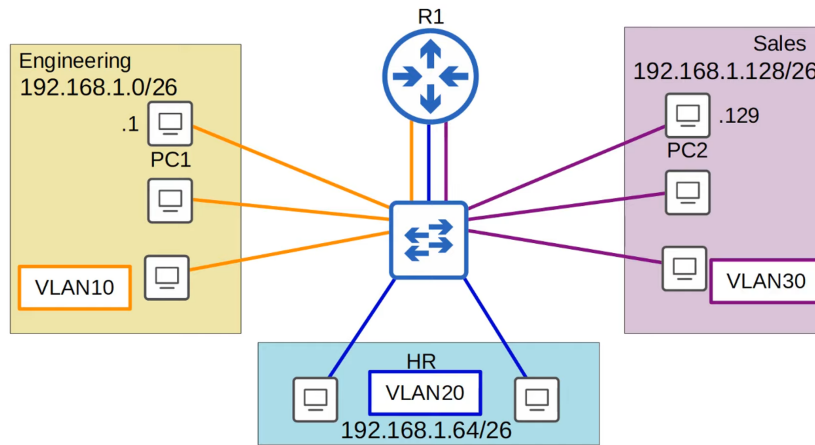
Network Performance :

- Reduce unnecessary BROADCAST traffic, which helps prevent network congestion, and improve network performance

Network Security :

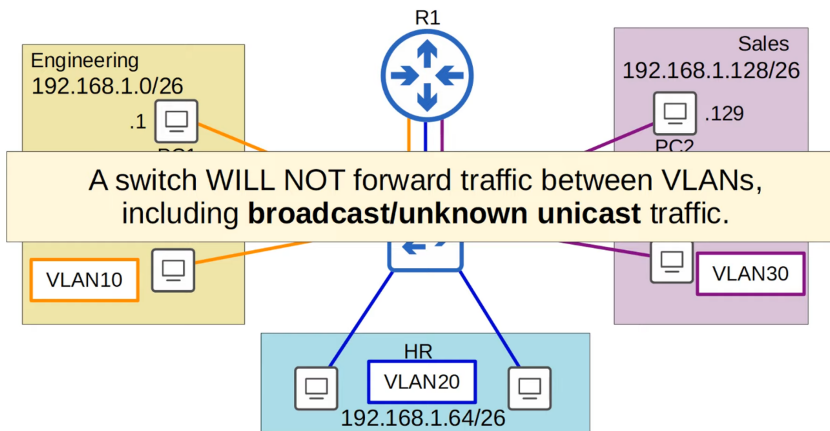
- Limiting BROADCAST and unknown UNICAST traffic, also improves network security, since messages won't be received by devices outside of the VLAN

What is a VLAN?

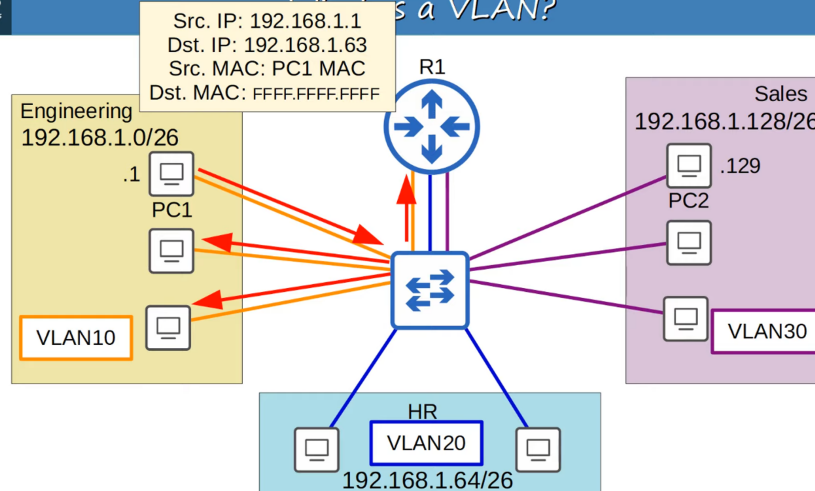


SWITCHES do not forward traffic directly between HOSTS in different VLANS

What is a VLAN?



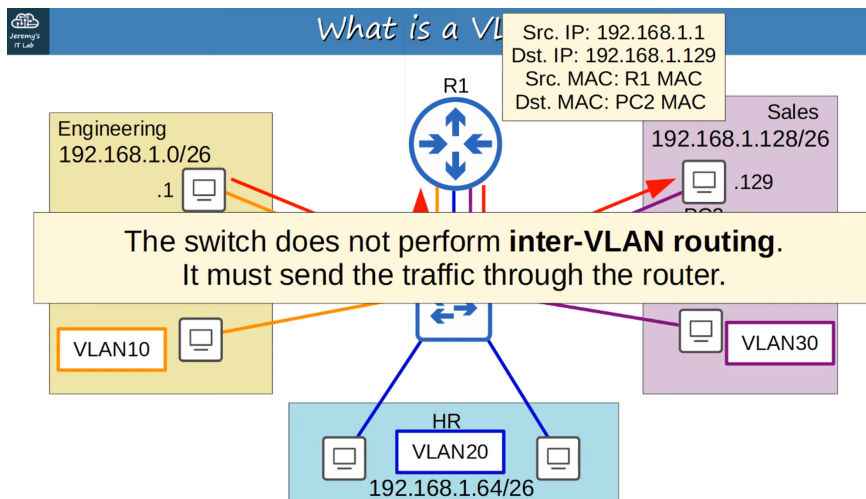
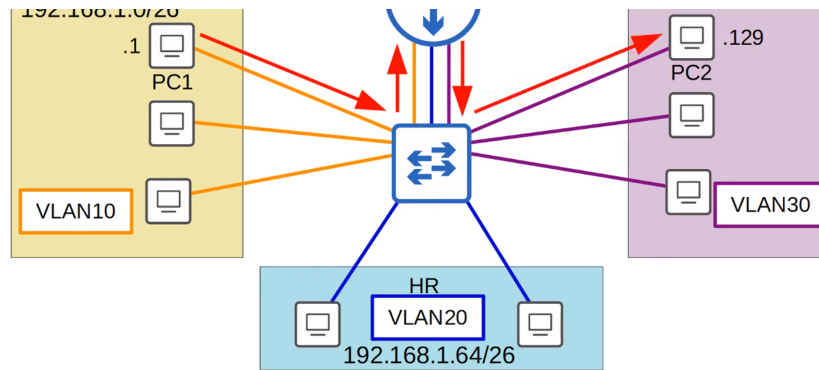
What is a VLAN?



Sending Packets to another VLAN (Routed through R1)

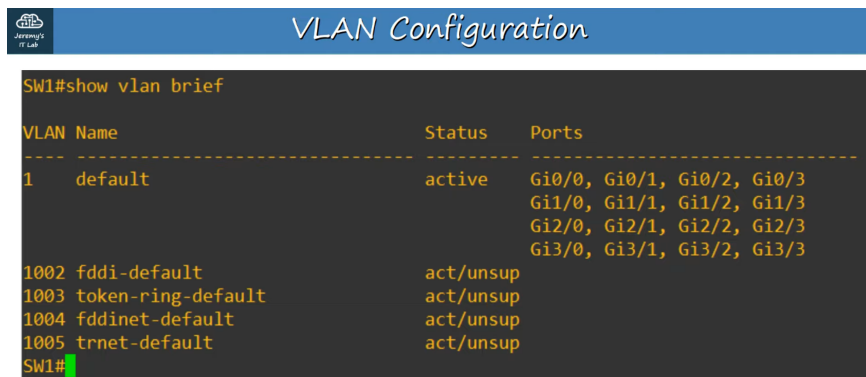
What is a VLAN?





HOW TO CONFIGURE VLANS ON CISCO SWITCHES

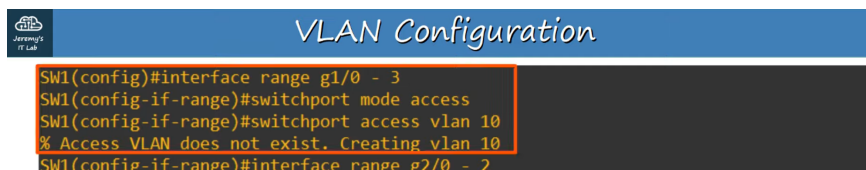
#show vlan brief



Shows which VLANs that exist on the SWITCH and what INTERFACES are in each VLAN

VLANs 1 (DEFAULT), 1002-1005 exist by default and cannot be deleted (5 VLANs)

HOW TO ASSIGN INTERFACES TO A VLAN



```
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
SW1(config-if-range)#interface range g3/0 - 3
SW1(config-if-range)#switchport mode access
SW1(config-if-range)#switchport access vlan 30
% Access VLAN does not exist. Creating vlan 30
SW1(config-if-range)#
```

1. Use the “interface range” command to select all the interfaces at once
2. Use the “switchport mode access” command to set the interface as an ACCESS PORT

WHAT IS AN ACCESS PORT?

- An ACCESS PORT is a SWITCHPORT which belongs to a single VLAN, and usually connects to end hosts like PCs.

SWITCHPORTS which carry multiple VLANs are called “TRUNK PORTS” (more info on TRUNK in next chapter)

3. Use the “switchport access” command to assign a VLAN to a PORT

VLAN Configuration

```
SW1(config)#do show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi0/1, Gi0/2, Gi0/3
                             Gi2/3
10   VLAN0010                active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
20   VLAN0020                active    Gi2/0, Gi2/1, Gi2/2
30   VLAN0030                active    Gi3/0, Gi3/1, Gi3/2, Gi3/3
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup

SW1(config)#vlan 10
SW1(config-vlan)#name ENGINEERING
SW1(config-vlan)#vlan 20
SW1(config-vlan)#name HR
SW1(config-vlan)#vlan 30
SW1(config-vlan)#name SALES
```

Use “#vlan <#>” to enter Configuration Mode for a given VLAN (this can also create a VLAN)

Use “#name ” to configure a NAME for your VLAN

To check your VLAN configuration, use “#show vlan brief”

VLAN Configuration

```
SW1(config)#do show vlan brief

VLAN Name                Status    Ports
-----
1    default                active    Gi0/0, Gi0/1, Gi0/2, Gi0/3
                             Gi2/3
10   ENGINEERING            active    Gi1/0, Gi1/1, Gi1/2, Gi1/3
20   HR                     active    Gi2/0, Gi2/1, Gi2/2
30   SALES                  active    Gi3/0, Gi3/1, Gi3/2, Gi3/3
1002 fddi-default          act/unsup
1003 token-ring-default    act/unsup
1004 fddinet-default       act/unsup
1005 trnet-default         act/unsup
SW1(config)#
```

Testing VLAN 10

Pinging from PC1 using 255.255.255.255 (FFFF:FFFF:FFFF) floods broadcast packets to R1 and VLAN10 hosts only

VLAN Configuration

